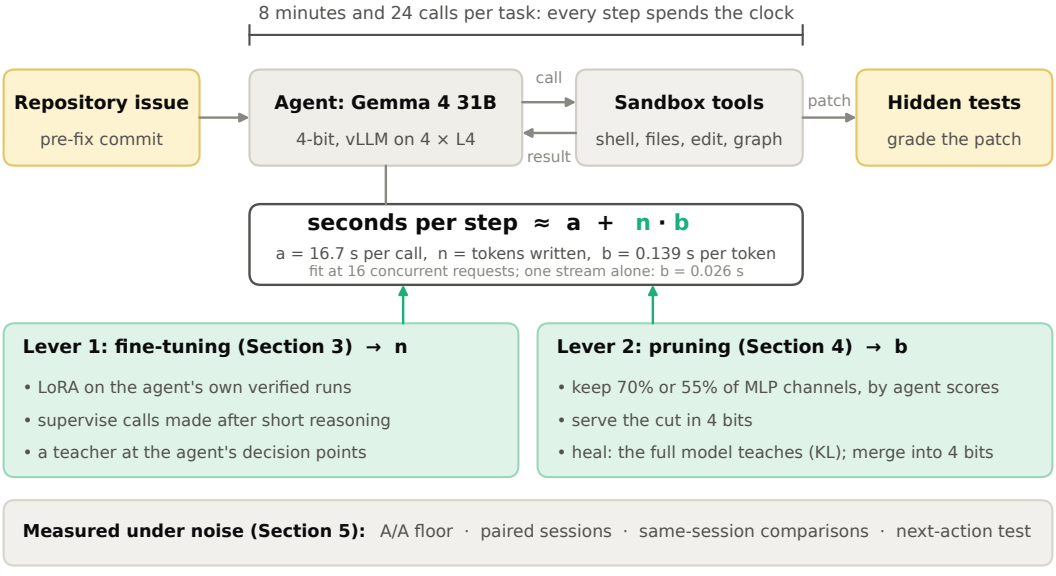
8 minutes and 24 calls per task: every step spends the clock
Repository issue
pre-fix commit
Agent: Gemma 4 31B
4-bit, vLLM on 4 × L4
call
result
Sandbox tools
shell, files, edit, graph
patch
Hidden tests
grade the patch
seconds per step ≈ a + n · b
a = 16.7 s per call, n = tokens written, b = 0.139 s per token
fit at 16 concurrent requests; one stream alone: b = 0.026 s
Lever 1: fine-tuning (Section 3) → n
• LoRA on the agent's own verified runs
• supervise calls made after short reasoning
• a teacher at the agent's decision points
Lever 2: pruning (Section 4) → b
• keep 70% or 55% of MLP channels, by agent scores
• serve the cut in 4 bits
• heal: the full model teaches (KL); merge into 4 bits
Measured under noise (Section 5): A/A floor · paired sessions · same-session comparisons · next-action test